

NeuroLearn Research Suite (NEURO_FOCUS)

Real-World PhysioNet EEG Technical & Research Project Report
A Modular Biomedical Signal Processing (BSP) Platform for EEG-Based Attention Quantification

Primary Dataset: PhysioNet_S001 (eegmmidb)	Methodology: 100% Non-ML / Explainable BSP
Sampling Rate (fs): 160.0 Hz	Channel Count: 64 Channels
Recording Duration: 61.0 Seconds	Real Quality Audit: 72.73% SQI Score

1. Executive Summary & Real-World Dataset Validation

This report presents an empirical biomedical signal processing study evaluating student cognitive state derived directly from the **PhysioNet EEG Motor Movement/Imagery Database (eegmmidb)**. The primary dataset consists of 64-channel EEG recordings sampled at 160.0 Hz from PhysioNet_S001 encompassing both resting baseline (Run 1) and active task conditions (Run 2).

Strict Non-ML Policy: In accordance with rigorous scientific standards, **zero machine learning or black-box predictive models** were used. All attention metrics were computed using zero-phase Butterworth bandpass filtering (1-40 Hz), 50 Hz powerline notch filtering, Welch Power Spectral Density (PSD), Hjorth parameters (Activity, Mobility, Complexity), and mathematical band ratio formulations (Beta/Theta).

2. Real-World Signal Quality & Preprocessing Audit

PhysioNet Dataset SQI Audit Score: 72.73%
Signal-to-Noise Ratio (SNR): Mean SNR = 12.74 dB
Excluded Channels: FP1, FP2
Preprocessing Settings: 1.0 - 40.0 Hz 4th Order Zero-Phase Butterworth Bandpass + 50.0 Hz Notch Filter + Linear Detrending + Z-Score Standardization.

3. Empirical Attention Analysis (Baseline vs. Task)

Metric	Run 1 (Baseline)	Run 2 (Task)	Scientific Significance
Average Attention Score	51.4 / 100	48.3 / 100	Mean cognitive score
Peak Attention Score	95.3 / 100	79.9 / 100	Maximum observed focused state
Minimum Attention Score	17.2 / 100	11.3 / 100	Nadir attention level
Attention Stability Index	79.8%	78.6%	Inverse standard deviation
Dominant Category	High Attention	High Attention	5-Level rule-based state

4. Statistical Hypothesis Testing & Bland-Altman Agreement

Hypothesis Test: Independent Two-Sample t-Test
Test Statistic: 0.494 (p-value = 0.6240)
Cohen's d Effect Size: d = 0.15
Bland-Altman Agreement: Mean Bias = 3.09 points, 95% Limits of Agreement = [-61.87, 68.05], Pearson r = -0.218.

5. Real-World Systematic Ablation Study Results

Systematic evaluation of Preprocessing Pipeline configurations executed directly on the PhysioNet dataset:

Preprocessing Pipeline	Mean Attention	Peak Attention	Stability Index	Dominant Category
Raw Signal	52.9 / 100	94.7	79.6%	High Attention
Bandpass Only	52.9 / 100	94.7	79.6%	High Attention
Bandpass + Notch	52.9 / 100	94.7	79.6%	High Attention
Full Preprocessing	51.4 / 100	95.3	79.8%	High Attention

6. Mathematical Equations & Academic References

- **4th Order Zero-Phase Butterworth Bandpass:** $|H(jw)|^2 = 1 / [1 + ((w^2 - w0^2)/(w*B))^{(2N)}]$
- **Hjorth Activity:** $\text{var}(x(t)) = \sigma_x'^2$ | **Mobility:** σ_x' / σ_x | **Complexity:** $\text{Mobility}(dx/dt) / \text{Mobility}(x(t))$
- **Welch PSD:** $\text{PSD}(f) = (1/K) \sum_{k=1}^K |\text{FFT}(x_k * w)|^2 / (L * U)$
- **Spectral Edge Frequency (SEF95):** $\text{Integral}_0^{\{f95\}} \text{PSD}(f) df = 0.95 * \text{Total_Power}$
- **Attention Ratio:** $\text{Beta} / \text{Theta} = \text{Power}(13\text{-}30\text{Hz}) / \text{Power}(4\text{-}8\text{Hz})$
- **Cohen's d:** $d = (\text{Mean}_1 - \text{Mean}_2) / s_{\text{pooled}}$

Academic References:

1. Goldberger, A. L., et al. (2000). 'PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals.' *Circulation*, 101(23), e215-e220.

2. Coelli, S., et al. (2018). 'EEG-based indices of attention during online learning tasks.' *IEEE Transactions on Biomedical Engineering*.

3. Hjorth, B. (1970). 'EEG analysis based on time domain properties.' *Electroencephalography and Clinical Neurophysiology*.